

Xueji Zou

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EDUCATION

University of Kansas

Bachelor of Science in Mechanical Engineering; GPA: 4.0/4.0

Lawrence, KS

Expected May 2029

PROJECTS

Flying-Wing UAV | Fusion 360, XFLR5, ANSYS Fluent

Summer 2026 – Present

- Modeled a 42 in flying-wing UAV in XFLR5 and Fusion 360; used iterative ANSYS CFD simulations to reduce drag 37.87% vs. baseline.
- Designed a modular 800 g airframe with a 250 g LW-PLA shell, carbon-fiber spars, and custom component mounts with threaded inserts.
- Selected a 2316 980KV motor and 3S 3200 mAh LiPo for a 45+ min endurance target; integrated Matek F405/iNav and GPS integration for autonomous flight.
- Conducted failure analysis on initial flight tests to identify CoG and launch instabilities; fabricated a PVC/bungee launcher for consistent, low-pitch launches.

Low-Speed Open-Circuit Wind Tunnel | SolidWorks, Arduino, C++, MATLAB

Spring 2026 – Present

- Designed and built a wind tunnel with a $9 \times 9 \times 18$ in test section, 3D-printed 5th-order polynomial contraction, honeycomb straightener, and dual mesh screens.
- Developed an Arduino-based DAQ system with a pitot-static tube and MPXV7002DP differential pressure sensor to measure test-section airspeed up to 32 mph.
- Designed and calibrated a 2-axis force balance using dual load cells and HX711 amplifiers to independently measure lift and drag across multiple angles of attack.
- Processed force data in MATLAB to compute and compare trends with ANSYS CFD predictions for a flying-wing UAV, analyzing discrepancies due to Reynolds-number mismatch and tunnel blockage.

Mousetrap Car | SolidWorks, Mechanical Design, Prototyping

Fall 2025

- Designed and built a vehicle capable of traveling along a narrow beam and stopping near a target location.
- Tested wheel alignment, axle configurations, and lever-arm lengths to improve stability and accuracy.

EXPERIENCE

Host & Cashier

JoyWok

February 2025 – Present

Overland Park, KS

- Greet and seat guests, manage the waitlist, and pace table assignments across days of up to 450 tickets and 1,300+ guests, including dinner rushes of up to 200 tickets.
- Process cashier transactions and takeout orders accurately, handling cash and card payments at the register.
- Resolve customer concerns and coordinate with servers and kitchen staff to ensure guests are seated, served, and checked out efficiently.

ACTIVITIES & HONORS

American Society of Mechanical Engineers (ASME) | University of Kansas

2025 – 2026

Chem-E-Car | University of Kansas

Fall 2025

Dean's List | University of Kansas

2025 – 2026

TECHNICAL SKILLS

CAD and Analysis: SolidWorks, Fusion 360, AutoCAD, XFLR5, ANSYS Fluent

Programming and Controls: MATLAB, C++, Arduino IDE, ESP32, iNAV

Additional: Microsoft Excel; English and Mandarin